HE3027: Housing Economics Project
Measuring Willingness to Pay for non-market amenity using Hedonic Regressions

1 Instructions

In this project, students, **in their groups of five**, will be valuing a non-market amenity by estimating hedonic house price regressions on a sample of housing transactions made in London, England. As highlighted in the lecture, hedonic model is an important economic tool for economists to measure the value of non-market amenities.

The non-market amenities of interest include:

- Crime
- Air Quality
- Accessibility to Tube stations

You are allowed to form your groups outside the tutorial groupings. I understand that some students may have difficulties doing so. I have created a google excel document to facilitate the matching process. You can click on this **link** to access the group list. Once you have your groups, please provide your group details in the document. For those groups that need additional group members, please list down your groups under the **incomplete tab** so others who are looking for project groups can link up. Please update the excel sheet as soon as possible to facilitate group formation. All students should have a group by the end of **week 4**. Please let me know if you face difficulties finding a group. To make sure that exchange students are integrated, and are able to learn more about Singapore, I also set a rule that there shouldn't be more than 2 (non-local) exchange students per group. [1]

[1] This rule does not apply to exchange students from Singapore. I understand that this could be a bit binding and I am happy to hear your suggestions.

The report should not be more 15 pages (double space font 12; excluding cover page and references but including tables and graphs) and must be submitted via **email**[2] on the **last day of Week 13** before 2359. This will take up **30% of the grade**.

The presentation is scheduled on **Week 12**. This will take up another **10% of the grade**. Each presentation should not take more than 15 minutes, with every member playing a role in the presentation. There will also be a peer review component to minimize free riding and this will be taken into account when deciding the final grade. **Groups will record their presentation via zoom and will send me the recording link (with permission to download) via email. There will be no classes on Week 12**

I also encourage students to reach out to me during office hours for consultation. Please drop me an email to arrange for an appointment.

2 Data Description

Students will be given a sample of housing transactions from the Land Registry database[3].

The dataset is land_registry_crime.dta for those who have chosen to work with crime, land_registry_pollution.dta for those who have chosen to work with pollution, land_registry_tube.dta for those who have chosen to work with transport accessibility.

Table 1 summarizes the variables within the dataset. This includes measures of neighbourhood quality, accessibility, amenities, local air quality and crime rates. Students will be working with the data provided and **will not be required** to collect their own data for this project. However, students are allowed to augment additional data into their analyses if they believe that these variables can help to improve their estimation. They could do so after learning how to

To facilitate the estimation of the hedonic house price regressions, a sample do-file will be provided to show how to conduct the estimation, export the results and graph the results. I will go through a demonstration during tutorial on STATA to ensure that everyone has the capacity to conduct these estimations.

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[3] This database basically records all the transactions that have taken place in England, Scotland and Wales.

3 Outline of Report

The following section describes a brief outline for the report. This serves as a rough guide. Students can include additional components if they deem fit but they must bear in mind the stipulated page limit.

- Select the variable of interest. In this project, the objective is to estimate the willingness to pay (WTP) for the variable of interest.
- Review the existing literature on estimating the WTP. Conduct a literature review (for at least 5 papers) associated with your variable of interests. You should discuss the following in your report:
  - why the non-market amenity of interest affects housing values? what is it capturing?
  - what are the potential challenges when estimating WTP for the non-market amenity of interest? What could be the reasons that the estimates could be biased?
  - How did the authors for each paper address the challenges with their empirical strategies?
  - While conducting the literature review, you should describe the setting of each study and the estimated effects as well (briefly).

  .
- Specify your regression model. Decide on the variables to include in your regression model. When selecting these variables, choose those that you believe could be correlated with your main variable of interest (either crime or pollution) and affect house prices. For instance, air quality could be worse off around industrial areas that are far away from the Central Business District (CBD). Without controlling for distance to CBD, one could over-estimate the WTP to avoid pollution as the estimates are capturing the discount in property prices in less accessible locations far from the city center.
- Conduct the estimation and interpret the coefficients surrounding the key estimates. There is no need to discuss all the variables that you have included. Only focus on those that are statistically significant and your key variable of interest. Consider sequentially adding in those variables and examine how it changes your key estimate (the WTP for your variable of interest). For instance, you can start off with housing characteristics (e.g leasehold, flat type), before adding in neighbourhood characteristics and locational characteristics. You can then consider adding in the location (e.g lsoacode, msoacode) and year fixed effects.

- Try to arrive with a specification with sensible results in terms of the size and the direction. For instance, if your estimates suggest that people are paying more to live in crime-ridden neighbourhoods, then something could be wrong! You may want to review your specification and the variables chosen.
- Take note that your estimates may not make sense due to the potential identification issues. Compare your results with the existing literature. Critically evaluate your empirical strategy and highlight potential problems (e.g omitted variable bias, reverse causality) that may exist in relation to the literature you have reviewed.
- List down several ways your estimated WTP measures could be useful. E.g Can we use these estimates to think about how much to spend on fighting crime? Relying on your WTP measures, compute the economic cost incurred by Londoners (from either crime or air pollution, depending on your variable of interest).

4 Relevant Literature

Some relevant papers associated with the different non-market amenities are provided below:

- Air Quality: Chay & Greenstone (2005), Bui & Mayer (2003), Currie *et al.* (2015)
- Crime: Gibbons (2004), Linden & Rockoff (2008), Pope & Pope (2012)
- Transport accessibility: Gibbons & Machin (2005), Baum-Snow & Kahn (2000), Billings (2011)

References

Baum-Snow, Nathaniel, & Kahn, Matthew E. 2000. The effects of new public projects to expand urban rail transit. *Journal of Public Economics*, **77**(2), 241–263.

Billings, Stephen B. 2011. Estimating the value of a new transit option. *Regional Science and Urban Economics*, **41**(6), 525–536.

Bui, Linda TM, & Mayer, Christopher J. 2003. Regulation and capitalization of environmental amenities: evidence from the toxic release inventory in Massachusetts. *Review of Economics and statistics*, **85**(3), 693–708.

Chay, Kenneth Y, & Greenstone, Michael. 2005. Does air quality matter? Evidence from the housing market. *Journal of Political Economy*, **113**(2), 376–424.

Currie, Janet, Davis, Lucas, Greenstone, Michael, & Walker, Reed. 2015. Environmental health risks and housing values: evidence from 1,600 toxic plant openings and closings. *American Economic Review*, **105**(2), 678–709.

Gibbons, Stephen, & Machin, Stephen. 2005. Valuing rail access using transport innovations. *Journal of urban Economics*, **57**(1), 148–169.

Gibbons, Steve. 2004. The costs of urban property crime. *The Economic Journal*, **114**(499), F441–F463.

Linden, Leigh, & Rockoff, Jonah E. 2008. Estimates of the impact of crime risk on property values from Megan's laws. *American Economic Review*, **98**(3), 1103–27.

Pope, Devin G, & Pope, Jaren C. 2012. Crime and property values: Evidence from the 1990s crime drop. *Regional Science and Urban Economics*, **42**(1-2), 177–188.

Table 1: Variables provided in the regression dataset

Variable Name	Description
price	Sale price stated on the transfer deed (in pounds)
postalcode	Postalcode
propertyty	D = Detached, S = Semi-Detached, T = Terraced, F = Flats/Maisonettes, O = Other
age	Y= newly built ; N = established residential building
duration	F = Freehold, L= Leasehold etc.
year	Year Property was sold
month	Month Property was sold
OA11CD	Output Area (OA) Code 2011
lsoacode	Lower Super Output Area (LSOA) 2011 Code
msoa11	Middle Super Output Area (LSOA) 2011 Code
transactiondates	transacted dates
quarter	Quarter Property was sold
yearquarter	year and quarter property was sold
bus_distnear	Distance from nearest bus stop (in meters)
thamesriv_dist	Distance from Thames River (in meters)
Bus_stopscount_200m	Counts of Bus stop in 200m buffer from property
percent_noedu	Percentage of residents with no education qualification
percent_minorities	Percentage of residents of minority race
lone_hh	Percentage of lone (single) parent households
detached_dum	=1 if property is a detached house
flat_dum	=1 if property is a flat
semi_d_dum	=1 if property is a semi-detached house
terrace_dum	=1 if property is a terrace house
freehold	=1 if property is under freehold
leasehold	=1 if property is under leasehold
heritage_count_200m	Counts of Grade 1 heritage buildings 200m from property
Nearest_Park_dist	Distance from nearest grade 1 park (in meters)
oslaua	Local Areas District (LAD) Code
grossannualpay	Average Gross Annual Pay for residents living in LAD in a year
hoursworked	Average number of hours worked (per week) for residents living in LAD in a year
jobdensity	Number of jobs per unit area in LAD in a year
totalpopulation	Population size in LAD in a year
unemployment	Unemployment rate in LAD in a year
pm10	average hourly particulate matter 10 from the nearest site (averaged across the
nox	average hourly nitrogen oxide from the nearest site (averaged across the year)
violent_crime	counts of all violent crimes in a year in LSOA
propertycrime	counts of all property crimes in a year in LSOA
allcrime	counts of all crimes in a year in LSOA
stn_postcode	"postcode for tube station"
stn_zone	"zone for tube station"
station	"tube station name"
station_km	"distance to the nearest tube station (km)"